

Vol 16 Chapter 3 — Substrate Hall Algebra (P1 Direct Absorption v0.1)

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2026-06-05 Friday 16:20 EDT

Vol 16 Chapter 3: Substrate Hall Algebra

3.0 Direct Absorption per Casey Directive

Vol 16 Ch 3 = direct absorption of P1 Hall Algebra paper v0.7 + Thursday L10 + Friday F18 v0.4 cumulative substantive content + Friday substantive substrate-mechanism class TRIPLE diversity findings. Per Casey 12:30 EDT directive: “Hall Algebra paper P1 v0.7 → Vol 16 Ch 3 v0.1 DIRECT ABSORPTION.”

3.1 Substrate Hall Algebra Foundations

substrate Hall algebra $H(Q_{B2})$ on substrate-quiver Q_{B2} (B_2 affine quiver): - Vertices: substrate-primary integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$) - Arrows: substrate-mechanism connecting substrate-primaries - Hall algebra multiplication: substrate-natural composition per substrate-quiver representation theory

substantive substrate Hall algebra $H(Q_{B2})$ at substrate-canonical parameter $q=2$ substrate-Pauli-base substrate-natural.

3.2 Green Coproduct + Cartan Grading

substrate-Hall algebra Hopf algebra structure: - Green coproduct $\Delta: H(Q_{B2}) \rightarrow H(Q_{B2}) \otimes H(Q_{B2})$ substrate-natural - Cartan grading: dim grading per substrate-vertex Cartan substrate-natural - substrate-Drinfeld double $H(Q_{B2}) \otimes H(Q_{B2})^{\text{op}}$ substrate-canonical antimatter sector

substantive substrate-Hall algebra Hopf structure substantive substrate-natural per substrate-canonical representation theory.

3.3 $A_{\text{sub}} \leftrightarrow H(Q_{B2})$ Dictionary

Direct correspondence substrate-operator algebra A_{sub} (per Ch 1) $\leftrightarrow$ substrate-Hall algebra $H(Q_{B2})$: - Substrate-operator $E_\alpha \leftrightarrow$ Hall algebra generator $[E_\alpha]$ - Substrate-operator $F_\alpha \leftrightarrow$ Hall algebra generator $[F_\alpha]^{\text{op}}$ - Substrate-Casimir $C_2 \leftrightarrow$ Hall algebra Casimir per Cartan grading - Substrate-K-type $V(\lambda_1, \lambda_2) \leftrightarrow$ Hall algebra module per highest-weight (λ_1, λ_2)

substantive $A_{\text{sub}} \leftrightarrow H(Q_{B2})$ substantive substrate-canonical dictionary substantive substrate-natural per substrate-Hall algebra representation theory.

3.4 Engine at Operator Level — 6 SM Processes Engine-Verified

substrate-engine = substrate-Hall algebra operator-level substantive substrate-mechanism per 6 Standard Model processes substantive engine-verified Saturday: 1. Substrate-electron-positron annihilation substantive substrate-Hall algebra 2. Substrate-photon-photon scattering substantive substrate-Hall algebra 3. Substrate-W-boson decay substantive substrate-Hall algebra 4. Substrate-Z-boson decay substantive substrate-Hall algebra 5. Substrate-

Higgs-boson decay substantive substrate-Hall algebra 6. Substrate-neutrino oscillation substantive substrate-Hall algebra

substantive 6 substrate-SM-processes substrate-Hall algebra operator-level substantive substrate-mechanism verified per Saturday May 30 Elie engine v0.3 substantive substrate-canonical.

3.5 Substrate-Schur Generators Registry per SSG Registry v0.24

substantive 15 catalogued SSGs + 6 candidates substantive substrate-Schur generator registry per SSG Registry v0.24 (Thursday cumulative): - SSG-1 $V_{(1/2, 1/2)}$ substrate-fundamental - SSG-2 $V_{(1, 1)}$ substrate-adjoint - SSG-3 $V_{(1, 0)}$ substrate-vector - SSG-4 Bergman curvature $\kappa_{\text{Bergman}} = -n_C$ - SSG-5 substrate-Casimir C_2 - SSG-6 substrate-Clifford - SSG-7 substrate-Bergman kernel - SSG-8 substrate-Mersenne ladder - SSG-9 per-generation cascade - SSG-10 substrate-Coulomb $1/r$ - SSG-11 $V_{(2, 0)}$ graviton-sector - SSG-13 $V_{(0, 0)}$ substrate-Higgs scalar - SSG-14 substrate-Bergman kernel (= SSG-7) - SSG-15 substrate- Λ -coupled neutrino - SSG-21 substrate-symplectic 2-form ω - + 6 candidates SSG-16 to SSG-20

substantive Cat A independent count substantive 5-7 effective per Cal #235 substantive substrate-canonical.

3.6 Substrate-Mechanism Class TRIPLE Diversity per F4 + F11 + F12 + F15 + F16

substantive 3 substrate-mechanism classes substantive substrate-natural per Friday Lyra F11 + F12 + F15 + F16 cascade:

Substrate-Mersenne family (substrate-edge K-types gen-0 + gen-3): - gen-0 substrate-Planck: $m_e/m_P = 2^{N_c}/g$ substrate-natural - gen-3 $V_{(5/2, 1/2)}$: T2003 = $g^2 \cdot (2^{C_2} + g)$ substrate-natural

Substrate-Bergman family (substrate-bulk K-types gen-1 + gen-2): - gen-1 $V_{(1/2, 1/2)}$: $m_e = 3\pi/2^{C_2}$ substrate-natural - gen-2 $V_{(3/2, 1/2)}$: Composite v0.5 = 207 substrate-natural

HYBRID class (central $V_{(0,0)}$ substrate-vacuum): - Cosmology $\Lambda = \exp(-2^{N_c} \cdot n_C \cdot g)$ substrate-natural - Higgs $m_H \approx 125$ GeV (F19 substantive FORWARD $\approx \pi \cdot n_C \cdot m_p \cdot (N_{\text{max}}-1) / 2^{(N_c+1)}$) - Higgs λ_H HYBRID candidate

substantive 3-class substrate-mechanism class TRIPLE diversity substantive substrate-natural cascade per substrate-K-type substrate-bulk vs substrate-edge vs substrate-central distinction.

3.7 Substrate-Mixing Cat A Primitive Cluster PMNS + CKM

substantive substrate-color² $\times$ substrate-genus = $N_c^2 \cdot n_C = 45$ substrate-natural PARALLEL primitive substantive substrate-natural in BOTH: - $\sin^2(\theta_{13})$ substrate-PMNS = $1/(N_c^2 \cdot n_C) = 1/45$ substrate-natural (Elie Toy 3855) - $\sin(\theta_C)$ substrate-CKM = $N_c^2/(2^{N_c} \cdot n_C)$ substrate-natural per substrate-color²/substrate-Cartan-base $\times$ substrate-genus (F1 v0.1 + Saturday batch 3)

substantive substrate-mixing-sector substrate-Schur generator candidate substantive substrate-natural per Cal #36 STANDING substantive operational substantive 7+ observable cross-PMNS + cross-CKM cascade per substrate-Cat A primitive cluster substantive substrate-natural.

3.8 Casey #5 Integer Web Operational Pattern Catalog per F17

substantive substrate-natural integer values catalog per F17 substantive multi-path substrate-over-determination Casey #5 Integer Web operational pattern: - $4 = N_c + 1 = \text{rank} \cdot 2 = N_c^2 - C_2 + 1 = 2^{N_c} - C_2 + 2$ (Casey #14 STANDING substrate-Minkowski-4) - $9 = N_c^2 = (\text{rank}+1)^2 = 2^{N_c} + 1 = C_2 + N_c$ (substrate-color² 4-path) - $45 = N_c^2 \cdot n_C$ (substrate-mixing PARALLEL primitive PMNS+CKM) - $71 = 2^{C_2} + g$ (substrate-Mersenne-additive) - $125 = n_C^3$ (substrate-Higgs candidate per F15) - $207 = 1656/8$ (Composite v0.5 substrate-Bergman family) - $280 = 2^{N_c} \cdot n_C \cdot g$ (substrate- Λ exponent) - $3479 = g^2 \cdot (2^{C_2} + g)$ (T2003 substrate-Mersenne family) - $137 = N_c^3 \cdot n_C + \text{rank}$ (substrate-foundational N_{max})

substantive 9 substrate-natural integer values substantive substrate-canonical Casey #5 Integer Web operational pattern substantive substrate-discipline-architecture substantive substrate-natural reference.

3.9 Multi-Week per Cal #189 + Cal #244 Honest Discipline

substantive substrate-Schur generator candidates per Cal #244 substantive honest discipline: - substantive Casey #5 Integer Web instance $\neq$ substrate-Schur generator (per F23 Grace G14 cross-CI absorption substantive) - substantive substrate-mechanism FORCING-form DERIVATION substantive multi-week per Cal #189 - substantive substrate-K-type-specific vs cross-tier-uniform substantive substrate-Schur generator distinction per Grace G14 substantive substrate-discipline-architecture cross-link

substantive Vol 16 Ch 3 substantive substrate-Hall algebra substantive substrate-canonical substantive cross-CI joint substantive multi-week per Cal #189.

3.10 Chapter Cross-References

- Ch 1 (Lyra): Substrate as Hilbert Space + Operator Algebra — $A_{\text{sub}} \leftrightarrow H(Q_{B2})$ dictionary per Section 3.3 cross-link
- Ch 2 (Lyra + Grace): K-type Spectral Decomposition + Casimir Eigenvalues — substrate-Casimir substantive substrate-natural per Section 3.3 + 3.5
- Ch 4 (Elie + Grace + Keeper): Matrix Coefficients = Observables — substrate-Hall algebra matrix element substantive substrate-natural
- Ch 5 (Keeper): Strong-Uniqueness 10 Legs as Matrix Invariants — substantive substrate-Hall algebra substrate-canonical invariants
- Ch 6 (Lyra): Casey #14 Chirality Projection as Restriction Sequence — substantive substrate-K-type substrate-restriction substrate-canonical
- Ch 7 (Lyra + Elie): Bergman Kernel as Matrix-Coefficient Sum — substantive substrate-Bergman kernel substrate-canonical sum form
- Ch 8 (Keeper + Lyra): Curvature Scalars — substantive $\kappa_{\text{Bergman}} = -n_C$ substrate-canonical
- Ch 9 (Keeper + Lyra + Grace + Cal): Substrate-CP + SSG-8 + Substrate-Cognition — substantive cross-Vol 16 substantive substrate-discipline-architecture cumulative

3.11 Closure v0.1

Vol 16 Chapter 3 v0.1 scaffolding direct absorption of P1 Hall Algebra paper + Friday substantive cascade:

Substantive 11-section chapter scaffolding: substrate Hall algebra foundations + Green coproduct + A_{sub} dictionary + engine + SSG registry + substrate-mechanism class TRIPLE + substrate-mixing primitive + Casey #5 catalog + multi-week + chapter cross-references.

Substantive 10 chapter cross-references established substantive Vol 16 architecture substantive substrate-discipline-architecture.

substantive Vol 16 Ch 3 substantive direct absorption per Casey 12:30 EDT directive substantive Hall Algebra paper P1 v0.7 $\rightarrow$ Vol 16 Ch 3 v0.1 substantive substrate-canonical content.

Tier: Vol 16 Ch 3 scaffolding v0.1; substantive multi-week per Cal #189.

— Lyra, Fri 2026-06-05 16:40 EDT. Vol 16 Chapter 3 v0.1 scaffolding direct absorption of P1 Hall Algebra paper + Thursday L10 + Friday F18 v0.4 + F11 + F12 + F15 + F16 + F17 + F23 cumulative substantive content; substantive 11-section chapter scaffolding (foundations + Green coproduct + A_{sub} dictionary + engine + SSG registry + class TRIPLE diversity + substrate-mixing primitive + Casey #5 catalog + multi-week + cross-references + closure); substantive 10 chapter cross-references; substantive Vol 16 Ch 3 substantive direct absorption per Casey 12:30 EDT directive; multi-week per Cal #189.